Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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SDC HACKATHON 2026

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1. Team & Project Overview

Team Name	Stewpid
Selected Domain	Domain 1: Aiversecity Future-Ready Workforce
Selected SDG Track (1–17)	SDG 4 (Quality Education) and SDG 8 (Decent Work and Economic Growth)
Project / Solution Name	Bridge - Reskilling Platform
Name (Leader)	Mahir Jabid Nafi
Email & Phone (Leader)	stewpeed.dev@gmail.com, (+60) 142411337
University Affiliation(s)	Taylor's University
Number of Team Members	1

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

People keep hearing that AI will change their jobs, but almost nobody tells them what to do about it. In LinkedIn's 2025 survey of more than 19,000 workers, 46% said learning AI feels like a second job. Randstad's 2025 Workmonitor found that only 48% of workers had been trained on the AI tools likely to change their work.

The people most affected are workers whose tasks are exposed to AI, fresh graduates entering a changing job market, and people thinking about switching careers. Many of them are also the least likely to train: the OECD reports that only about 20% of low-skilled adults take part in job-related training, against more than 40% of medium and high-skilled adults.

Career sites and job portals don't fix this. They list hundreds of roles and courses, but they don't say which ones fit this person, or how much of a new skill they already have. So every new skill looks completely new, and people freeze or give up before they start. A graphic designer looking at video editing doesn't see that colour, composition and photo retouching already carry over.

This matters for SDG 4 and SDG 8. If workers can't see a realistic next step, skill gaps grow, people stay in roles that are shrinking, and the benefits of AI go mostly to those who were already ahead.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Bridge is a web app that tells a person which skill is worth learning next, and how much of it they already know.

The user starts by uploading a CV or answering a short questionnaire about their work, tools, education and

interests. Bridge pulls out their skills and matches each one to ESCO, the European Commission's official list of jobs and skills. It matches by meaning, not exact words, using a sentence-embedding model (BAAI bge-small-en-v1.5), so "photo retouching" is recognised as "edit photographs".

Bridge then scores 117 roles. Each score has four parts the user can open up: how much of the role they already know, how quickly they could close the gap at their weekly hours, future demand, and how well it fits the kind of work they enjoy. Demand comes from updatable sources, each shown with its date: live job posting counts, O*NET Bright Outlook and the ILO 2025 index of exposure to generative AI. It is shown as two labelled estimates, moderate and fast AI change. Every role is also tagged with the UN SDGs it serves, with a reason.

Choosing a role turns the gap into a pathway. It starts from the topics closest to what the person knows, with time estimates and real, hand-checked courses (96 of the 164 are free).

Learners can share their progress with an employer through a short code, and universities can follow how their students grow, with the students' consent.

Generative AI is kept out of the scores, so it can't invent jobs, courses or numbers. From the workshops, I used cloud hosting (the app runs on Replit), database design (SQLite built from public data), GPU computing (RunPod, to embed the ESCO data) and a GenAI API (Claude, as an optional CV reader).

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Frontend: React 18 and React Router, built with Vite. Plain CSS with design tokens.

Backend: Python with FastAPI and Uvicorn. One server runs the API and serves the built app.

Database: SQLite, built from public data (ESCO, the ESCO to O*NET crosswalk, O*NET, the ILO index and dated job-posting snapshots). A second small SQLite file stores a share code only when a learner asks for one.

AI: BAAI/bge-small-en-v1.5 sentence embeddings. The ESCO skills and job titles were embedded once on a RunPod GPU; the user's own phrases are embedded on the server's CPU with the same model. Optional CV reading with Claude through Replit AI Integrations (off in the live demo).

CV reading: pypdf and python-docx. The file is read in memory and never saved.

External APIs: ESCO API, Remotive and Arbeitnow job APIs (trend refresh), Open Library (book search).

How it connects:

Browser (React) → FastAPI on Replit → SQLite (roles, skills, demand) + embeddings → matches, pathway and courses back to the browser.

Data refresh: ESCO, O*NET, ILO and job APIs → data scripts → SQLite, with the date of each snapshot.

One-off GPU job: ESCO text → RunPod GPU → embeddings file → committed to the repo and loaded by the server.

Sponsor resources: Yes. Replit hosts the live prototype (https://bridge-reskill.replit.app). RunPod (one NVIDIA A40 pod) ran the job that embedded 3,943 ESCO skills and job titles. Replit AI Integrations (Claude) is built in as an optional CV reader.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

I started with the people: workers worried about AI, graduates and career switchers. The research pointed to fear more than a lack of information. A new skill feels completely new, so people don't start. That became the core idea: show people how much they already know.

I designed every screen in Figma first, then built each flow and tested it in the browser, on desktop and on phone sizes. Each score can be opened up into its four parts, because a number without a reason doesn't help anyone decide. A comfort question lets people choose how far they want to stretch, so nobody is pushed into a big jump they don't want. Every CV answer is shown for the user to check before it is used.

To keep it workable, the matching uses public data and simple maths that run on a normal CPU. The only GPU step ran once on RunPod. The app is already live on Replit as one server. Phase 2 is about adding Malaysian data (local job postings and HRD Corp claimable courses), Bahasa Melayu and Chinese, and testing with real users.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	SDG 4.4: by 2030, substantially increase the number of youth and adults who have relevant skills, including technical and vocational skills, for employment, decent jobs and entrepreneurship. SDG 8.5: by 2030, achieve full and productive employment and decent work for all women and men, including young people.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Only 48% of workers have been trained on the AI tools likely to change their jobs (Randstad Workmonitor 2025, 27,062 workers in 35 markets). 46% say learning AI feels like a second job (LinkedIn research by Censuswide, August 2025, 19,000+ workers in 14 countries). About 20% of low-skilled adults take part in job-related training, against more than 40% of medium and high-skilled adults (OECD, 2019, Getting Skills Right: Future-Ready Adult Learning Systems).
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	1. Start rate: % of users who start the first topic of a pathway after seeing their matches. 2. Head start: % of a chosen role's skills the user already has, which they don't need to learn from zero (Bridge already calculates this for every role).
Phase 2 Impact Target	In a test with at least 20 people (students and career

Metric Category	Your Response
Your realistic, percentage-based target by the final prototype presentation *e.g., "20% improvement in computational resource use."*	switchers): 60% start a first topic in the same session, and on average the chosen pathway is at least 25% shorter than learning the role from zero, because skills they already have are counted.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☒ No

If yes, briefly describe what was pre-existing and which team member(s) owned it: None.

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☒ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
React, React Router, Vite, jsPDF	Code	MIT
FastAPI, Uvicorn, NumPy, pypdf, python-docx	Code	MIT / BSD-3-Clause
fastembed, python-multipart	Code	Apache 2.0
Anthropic Python SDK	Code	MIT
BAAI/bge-small-en-v1.5	Model	MIT
Claude (optional CV reading, via Replit AI Integrations)	Model	Anthropic commercial terms
ESCO occupations, skills and ESCO to O*NET crosswalk	Dataset	Free reuse with attribution (European Commission)

O*NET 31.0 interest types and Bright Outlook	Dataset	CC BY 4.0
ILO 2025 GenAI exposure index (Gmyrek et al.)	Dataset	CC BY 4.0
Remotive and Arbeitnow job posting counts	Dataset	Free public API terms
Course links (OpenLearn, MIT OCW, Kaggle, Coursera and others)	Dataset	Links and facts only
Fonts: Bricolage Grotesque, JetBrains Mono, Switzer	Code (fonts)	SIL OFL 1.1 / ITF Free Font License
3D icons (thiings.co), assistant logos (Simple Icons)	Design assets	Free to use / CC0 1.0

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☒ No

If yes, please describe the nature of the agreement and which member it concerns: None.

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



Date: 25 October 2026